Ali Muhammad Asad

+92 333 2103150 | alimuhammadasadlm10@gmail.com | AliMuhammadAsad | ali-muhammad-asad | Karachi, Pakistan

Education

Habib University — Karachi, Pakistan
B.Sc. in Computer Science; ***CGPA: 3.71 / 4.00*** *(80% 4-Year Academic Merit Scholarship)* — *Aug 2021 – Jun 2025*

- Honors: **Graduate School Curation Program (GSCP)** Fellow; Top 10 in regional round, qualified for **ICPC Pakistan National Finals** (2023); **Stanford IHP** (50% merit grant).
- **Relevant Coursework:** Deep Learning, Large Language Models, Computational Intelligence, Reinforcement Learning, Probabilistic Graphical Models, Probability & Statistics, Statistical Inference, Algorithms, Nature of Computation, Computer Architecture, OS, Database Systems.

Alpha College — Karachi, Pakistan
GCE A Levels in Pre-Engineering; ***2A*, 1A, 1B*** *(100% Academic Merit Scholarship)* — *Aug 2019 – Jun 2021*

Publications & Preprints

Pashto Poetry Attribution using Deep Learning Techniques | **[IEEE Xplore]** — *IEEE HONET 2025*

- **First Author** · Curated the first publicly available Pashto poetry dataset (35,637 couplets, 35 poets); engineered data augmentation pipelines (random deletion/swap) and evaluated Random Forest and Bi-LSTM models, achieving **85.25% classification accuracy**.

UrduLM (Project ALIF): A Resource-Efficient Monolingual Urdu Language Model | **[arXiv:2601.17664]**
Preprint 2026

- **Co-Author** · Developed the custom regex-BPE tokenizer and automated data collection pipelines for a 100M-parameter Urdu decoder LM pre-trained on a curated 33GB corpus, achieving competitive performance against multilingual models up to 30× its size.

Selected Research Experience

ALIF: Urdu Generative Large Language Model | ***Project Hub*** — Habib University
Capstone Contributor (Tokenizer & Data Collection) · Preprint: arXiv:2601.17664 — *Aug 2024 – May 2025*

- Engineered automated web-scraping and data collection pipelines aggregating a 33 GB monolingual Urdu corpus (∼13M rows, ∼5–6B tokens) across news archives, web crawls, and digitized literature.
- Independently designed and built a custom regex-BPE tokenizer (32k vocab) yielding **∼8% fewer tokens than GPT-4o** (`o200k_base`: 555 vs. 605 tokens/500 words) and **∼30% fewer than multilingual tokenizers** (mBERT, LLaMA-2: 800+ tokens).
- Collaborated on multi-GPU pre-training with PyTorch and Hugging Face Transformers using DDP/FSDP across **4× NVIDIA V100 GPUs (32GB)**; scaled architecture to an exploratory **1B model** with 0.5M token batches via gradient accumulation.
- Evaluated downstream few-shot benchmarks: on the capstone evaluation suite, the team's Alif-1B achieved **78.5% sentiment accuracy** (vs. 50% Qwen 2.5 3B), **51.2 BLEU in grammar correction** (vs. 39.8 LLaMA 3.2 3B), and a 78% win rate in LLM-as-a-judge tests (Gemini 2.0 Flash).

Empirical Performance Benchmarking of Graph Databases | ***GitHub*** — Habib University
Undergraduate Researcher · Advised by Dr. Qasim Pasta · Summer Research Program — *Jun 2024 – Aug 2024*

- Conducted an empirical scalability study evaluating four graph database engines under the LDBC Social Network Benchmark (SNB): **Neo4j**, **Memgraph** (in-memory C++), **Apache HugeGraph**, and **RedisGraph** (now end-of-life).
- Benchmarked graph datasets across scale factors **SF-0.1, SF-1, and SF-10** (up to 3 GB graphs), profiling ingestion throughput, multi-hop path query latency, and resource footprint across hardware constraints.
- Found that Memgraph's query latency grew significantly more slowly with dataset size; on slowest interactive complex queries at 3 GB scale, Memgraph completed in **3.8s vs. 12.0s for Neo4j** (> 3× speedup), while data loading time scaled linearly with dataset size.

Multi-Author Writing Style Analysis | ***GitHub*** — Habib University
Course Research Project · Evaluated on CLEF PAN 2025 Benchmark — *Sep 2024 – Dec 2024*

- Investigated style change detection and authorial transition boundary classification on multi-author Reddit documents from the international PAN 2025 benchmark dataset.
- Engineered n-gram (unigram/bigram) and lexical feature extractors with class-weighted logistic regression, achieving validation F1-scores of **0.791** (Easy) and **0.666** (Medium) on the PAN 2025 validation set, substantially surpassing official baselines (0.414 and 0.506).

Systems & Computational Projects

5-Stage Pipelined RISC-V Processor Design | *GitHub* *Computer Architecture*

- Implemented a 32-bit single-cycle processor and extended it into a **5-stage pipelined RISC-V processor** in Verilog HDL.
- Engineered hardware hazard detection, data forwarding units, and branch prediction logic; verified execution of sorting algorithms (bubble sort) in Vivado simulation.

Bayesian Networks for CRISPR-Cas13 Guide Efficiency | *GitHub* *Probabilistic Graphical Models*

- Modeled guide RNA cleavage efficiency for CRISPR-Cas13 systems in Python using `pgmpy`; applied Hill Climbing for DAG structural learning to capture sequence motif dependencies.

Teaching & Academic Appointments

Habib University Karachi, Pakistan
Teaching Assistant & Undergraduate Course Developer *Jan 2022 – May 2025*

- **Teaching Assistant (12 Course Appointments):** Served as TA across 12 CS and Math courses: Algorithms: Design & Analysis, Data Structures I & II, Algorithmic Problem Solving, Nature of Computation (Theory of Automata), Discrete Math (3 terms), OOP, Graph Data Science, Competitive Programming, and Calculus II; mentored 200+ students through weekly tutorials and recitations.
- **Curriculum Authoring (Summer Research Program):** Collaborated with CS faculty to author the curriculum, syllabus, and algorithmic problem sets for a new university course in *Competitive Programming* (now actively taught in the official CS curriculum).

Professional Experience

Habib University Karachi, Pakistan
Data Associate | Office of the Chief Operating Officer (COO) *Feb 2026 – Present*

- **Centralized Institutional Data Platform:** Migrated university-wide institutional databases (2018–present, 2,400+ students and alumni) from SQL Server to PostgreSQL; developed 15–20 interactive dashboards and internal portals in Next.js and React across 40–50 relational views, centralizing cross-departmental reporting for university leadership.
- **Advancement Allocation Engine:** Partnered with University Advancement on gift operations tracking ∼2,800 international donors and ∼5,000 gift records (PK, US, CA, UK); architected an algorithmic allocation engine matching 600–700 financial aid recipients annually to multi-criteria endowments (academic merit, financial need, degree programs), compressing allocation cycles from weeks to 1 day.
- **Institutional Governance & Accreditation:** Co-authored 2 university-wide AI governance policies regulating generative AI usage across coursework and administration; conducted Course and Program Learning Outcome (CLO-PLO) curriculum mappings for National Computing Education Accreditation Council (NCEAC) accreditation.

Habib University Karachi, Pakistan
Management Trainee | Office of Institutional Research (OIR) *Jul 2025 – Jan 2026*

- **Production Multi-Agent RAG Platform:** Architected a conversational analytics platform in Python, FastAPI, LangGraph, and Gemini with MongoDB Atlas Vector Search and Redis caching (< 3s response latency), secured via role-based access control (RBAC); compressed institutional analysis cycles for Deans and Program Directors from days to under 1 hour via natural language queries.
- **Full-Stack Deployment & Automation:** Built and deployed containerized microservices using Docker, AWS, and Next.js/React frontends; automated institutional ETL pipelines for course evaluations, faculty teaching loads, and historical student performance metrics.

Technical Skills & Certifications

Programming Languages: Python, SQL, Cypher (Graph Query), Verilog HDL, JavaScript/TypeScript

Machine Learning & AI: PyTorch, Hugging Face Transformers, Distributed Training (DDP, FSDP), Tokenization (BPE), Bi-LSTM, Random Forest, Large Language Models (LLMs), LangGraph, Multi-Agent RAG, Bayesian Networks (`pgmpy`)

Data Systems & Databases: PostgreSQL, MongoDB Atlas (Vector Search), Redis, Neo4j, Memgraph (in-memory C++), LDBC SNB Benchmarking

Systems & Frameworks: Linux/Unix CLI, Docker, AWS, Git, Vivado, FastAPI, Next.js, React

Certifications: CITI Responsible Conduct of Research, Neo4j Certified Professional, Neo4j Graph Data Science